Homework – Multi-Level Scenario Analysis and Process Transformation

Task 1: Defect and Process Identification

The table below classifies each observed practice under its dominant Level 1 symptom.

	Observed Practice	Level 1 Symptom
1	Developers implement client feature requests right away without checking for conflicts with existing code.	Chaotic Process
2	Project managers set timelines from personal "gut feeling" instead of recorded historical data.	Chaotic Process
3	Each developer keeps a personal folder structure and naming convention.	Lack of Standardization
4	Subcontractors are brought in on short notice with no formal check on their deliverables.	Chaotic Process
5	Testing is a quick manual walkthrough done by the developer just before deployment.	Reactive Management

Justification

1. There is no requirements-review or impact-analysis step before a change is made, so each request is handled purely on the developer's own initiative rather than through a repeatable procedure. This is the definition of a chaotic process: work proceeds without a formal method, and the outcome depends on whoever happens to touch the code.

2. Estimating from instinct rather than tracked project metrics is also a chaotic trait there is no consistent, repeatable method for planning, so every estimate is really an individual's private judgment call rather than an organizational practice.

3. When conventions differ from developer to developer, the knowledge needed to maintain the code lives only in each person's head. Once that person leaves, no one else can safely work on their part of the system. a textbook lack-of-standardization problem.

4. Bringing on outside vendors without any agreed acceptance criteria means their work quality is left entirely to chance. Nothing in the process forces a check before their output is accepted, which is again a symptom of chaotic, undefined process control.

5. Testing that happens only in the final moments before release is classic reactive management: quality is addressed at the last possible point, so problems are found (if at all) only once it is nearly too late to fix them properly.

Task 2: Level 2 (Managed) Remediation Plan

Five project-level controls are proposed below to move EduTech Solutions toward CMMI Level 2, each linked to a specific Process Area.

	Identified Failure	Level 2 Process Area	QA Mechanism
1	Features are built without checking their effect on the existing system.	REQM – Requirements Management	Introduce a Change Request form: every new request is logged, checked against a requirements/traceability record, and approved before development starts.
2	Schedules are set from personal judgment with no supporting data.	PP – Project Planning	Maintain a Project Plan built from a simple estimation worksheet that records scope, effort, and past project data instead of relying on guesswork.
3	Folder and naming conventions differ across developers.	CM – Configuration Management	Adopt one Configuration Management Plan describing the standard repository layout and naming rules, with a short checklist used before code is committed.
4	Subcontractor deliverables are accepted without any quality review.	SAM – Supplier Agreement Management	Require a signed Supplier Agreement plus a deliverable acceptance checklist that a subcontractor's work must pass before it is integrated.
5	Testing is a rushed manual pass right before release.	PPQA – Process and Product Quality Assurance	Set up a written Test Plan with defined test cases and a QA sign-off log confirming testing was completed and reviewed before deployment.

Conclusion:
Level 1 organizations work in a chaotic and reactive way. Level 2 introduces basic project-level controls and repeatable processes. Level 3 standardizes processes across the organization. QA helps improve quality through documentation, reviews, checklists and defect tracking.